

OPERATING SYSTEM

Important Examination Questions

Pattern: 3×2 -Mark Questions + 2×16 -Mark Questions for Each Topic

Topics Covered: Introduction, Process Management, CPU Scheduling, Process Synchronization, Memory Management and File System.

1. Introduction to Operating System

Important 2-Mark Questions

- 1. Define an Operating System. State its main objectives.
- 2. What is a system call? Give any two examples.
- 3. Differentiate between kernel and shell.

Important 16-Mark Questions

- 1. Explain the functions and services of an Operating System in detail.
- 2. Explain the different types of Operating Systems with suitable examples.

2. Process Management

Important 2-Mark Questions

- 1. Define a process. What is a Process Control Block (PCB)?
- 2. List the different states of a process.
- 3. What is a context switch?

Important 16-Mark Questions

- 1. Explain process states and the Process State Transition Diagram in detail.
- 2. Explain Process Control Block (PCB), context switching and inter-process communication (IPC) in detail.

3. CPU Scheduling

Important 2-Mark Questions

- 1. Define CPU scheduling. What is a ready queue?
- 2. Differentiate between preemptive and non-preemptive scheduling.
- 3. Define turnaround time and waiting time.

Important 16-Mark Questions

- 1. Explain FCFS, SJF, SRTF, Priority and Round Robin scheduling algorithms with examples. Compare their advantages and disadvantages.
- 2. Solve a suitable CPU scheduling problem using FCFS, SJF/SRTF, Priority and Round Robin, draw the Gantt chart, and calculate average waiting time and average turnaround time.

4. Process Synchronization

Important 2-Mark Questions

- 1. What is a race condition?
- 2. Define critical section. State the requirements for a correct critical-section solution.
- 3. What is a semaphore? Differentiate between binary and counting semaphores.

Important 16-Mark Questions

- 1. Explain process synchronization and the critical-section problem. Discuss mutex locks, semaphores and monitors.
- 2. Explain the classical synchronization problems: Producer–Consumer, Readers–Writers and Dining Philosophers.

5. Memory Management

Important 2-Mark Questions

- 1. What is paging?
- 2. Differentiate between internal and external fragmentation.
- 3. What is a page fault?

Important 16-Mark Questions

- 1. Explain paging, page tables, demand paging and virtual memory in detail.
- 2. Explain FIFO, Optimal and LRU page-replacement algorithms with a suitable reference-string example and calculate the number of page faults.

6. File System

Important 2-Mark Questions

- 1. What is a file? List any four file attributes.
- 2. List the basic file operations.
- 3. Differentiate between contiguous, linked and indexed file allocation.

Important 16-Mark Questions

- 1. Explain file-system structure, file operations, access methods and directory structures in detail.
- 2. Explain contiguous, linked and indexed file-allocation methods and discuss their advantages and disadvantages.

Most Important Questions – Last-Minute Revision

- 1. Explain OS functions, services and types of Operating Systems.
- 2. Explain process states, PCB and context switching.
- 3. Explain CPU scheduling algorithms and solve numerical problems using Gantt charts.
- 4. Explain process synchronization, critical section, mutex and semaphores.
- 5. Explain paging, virtual memory and page-replacement algorithms.
- 6. Explain file allocation methods and directory structures.
- 7. Practice CPU scheduling numerical problems: FCFS, SJF, SRTF, Priority and Round Robin.
- 8. Practice page-replacement numerical problems: FIFO, Optimal and LRU.

Exam Strategy: For 16-mark answers, write a definition/introduction, neat diagram or algorithm, detailed explanation, example, advantages/disadvantages, and conclusion wherever applicable. For numerical problems, show the Gantt chart and all calculations clearly.